

Bioinformatics Analysis Workflow



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Processing of raw sequencing data

Third-generation data:

The third-generation sequencing instrument PacBio, which utilizes single-molecule sequencing technology, has the advantage of extremely long reads and plays an important role in the assembly of complete genomes. The raw sub-level data is saved in the BAM format file, which includes sequencing sequences, base quality values, and other information. By performing data statistics on the raw data, information such as the total amount of data, read lengths, and quality value distribution can be obtained.

Second-generation data:

The raw data obtained from sequencing (Raw Data) may contain a certain proportion of low-quality data. In order to ensure the accuracy and reliability of subsequent information analysis results, the raw data needs to be filtered to obtain clean data (Clean Data). The specific processing steps are as follows:

Remove reads that have a proportion of low-quality bases (quality value ≤ 38) exceeding a certain threshold (default set to 40bp).

Remove reads that have a proportion of N bases exceeding a certain threshold (default set to 10bp).

Remove reads that have an overlap with the adapter exceeding a certain threshold (default set to 15bp).

For projects with small genomes or potential host contamination, the reads that may originate from the host should be filtered out by aligning them with the host database.

The Canu software (version: 2.0) is used to assemble the genome using third-generation reads, which produces preliminary assembly results that reflect the basic genomic information of the sample. Then, the Racon software (version: 1.4.13) is used for three rounds of error correction based on the third-generation sequencing data to correct the assembly results. After that, three rounds of error correction using second-generation reads are performed using the Pilon software (version: 1.22) to obtain the final assembly results.

Genome Component Analysis

Coding Genes

Starting from the final assembly results (≥ 500 bp) of each sample, ORF (Open Reading Frame) prediction and filtering are performed using the corresponding software. The results can be found in "03.Genome_Component.tar.gz" or "03.Genome_Component/Gene".

For samples such as bacteria and archaea, the GeneMarkS software (version 4.17) (<http://topaz.gatech.edu/>) is used for coding gene prediction and filtering.

Repeat Sequence Analysis

Based on the distribution of repetitive sequences in the genome, they can be divided into two major categories: dispersed repeat sequences and tandem repeat sequences. Dispersed repeat sequences are a type of repeat sequence that differs in organization from tandem repeat sequences, and they are scattered throughout the genome. Tandem repeat sequences (TR) are repetitive sequences that consist of adjacent, repeated patterns of a specific nucleotide sequence, repeated two or more times. The results can be found in

"03.Genome_Component.tar.gz" or "03.Genome_Component/Repeat".

The prediction of dispersed repeat sequences is performed using the RepeatMasker software (version 4.0.5), while the TRF (Tandem Repeats Finder, version 4.07b) is used to search for tandem repeat sequences in the DNA sequences.

Non-coding RNA (ncRNA)

Non-coding RNA (ncRNA) is a class of RNA molecules that are widely present in small genomes, bacteria, archaea, and eukaryotes, and they perform various biological functions. Unlike coding RNA, ncRNA does not carry information for protein translation and directly acts on biological processes at the RNA level. For microorganisms, common types of ncRNA include sRNA, rRNA, tRNA, snRNA, snoRNA, and miRNA, among others. The results can be found in "03.Genome_Component.tar.gz" or "03.Genome_Component/ncRNA".

Common methods for predicting different types of ncRNA are as follows:

tRNA: The tRNAscan-SE software can be used to predict tRNA regions and their secondary structures.

rRNA: By aligning against an rRNA database, rRNAs that are closely related to the database (with a default identity of $\geq 50\%$) can be identified. Additionally, the rRNAmmer software can be used to predict newly identified and unannotated rRNAs, combining both results.

sRNA: First, the Rfam software is used for Rfam database annotation. Then, the cmsearch program (version 1.1rc4) is used to determine the final sRNA predictions.

Genomics Islands

Genomic Islands (GIs) are parts of the genome in bacteria, phages, or plasmids that show evidence of horizontal origin. The sizes of GIs usually range from 10 to 500 Kbp, and their GC content and codon usage differ from the rest of the

genome. Many subcategories of GIs are based on their functional characteristics, such as Pathogenicity Islands (PAIs) that contain virulence genes related to pathogenic mechanisms, and Antibiotic Resistance Islands that contain numerous antibiotic resistance genes. The results can be found in "03.Genome_Component.tar.gz" or in the Genomic_Island folder within 03.Genome_Component.

IslandPath-DIOMB software is used for predicting genomic islands by detecting nucleotide biases (phylogenetically biased) and mobility genes (such as transposases or integrases) in the sequences to determine the presence of GIs and potential horizontal gene transfer.

Transposons

Transposons (abbreviated as Tn), also known as transposable elements, are basic units that exist on chromosomal DNA and have the ability to replicate and move autonomously. Transposition, or DNA transposition, is a genetic rearrangement phenomenon mediated by transposable elements and occurs at a much lower frequency compared to homologous recombination. Transposons can transfer between different replicons and insert from one site to another in a non-canonical recombination manner, resulting in various genetic effects on the structure and expression of genes at the new site.

TransposonPSI software is used to predict transposons. It searches for potential transposon sequences in genomes or protein sequences through PSI-Blast homology alignment. The results can be found in "03.Genome_Component.tar.gz" or in the Transposon folder within 03.Genome_Component.

Prophage

The nucleic acid integrated into the host genome of a temperate phage is referred to as a prophage. Prophage prediction is performed using the phiSpy



tool (Nov11_v2.3), which is a highly accurate and precise prediction tool. It supports GenBank format files. The results can be found in "03.Genome_Component.tar.gz" or in the Prophage folder within 03.Genome_Component.

CRISPR

CRISPR (Clustered Regularly Interspaced Short Palindromic Repeat Sequences) is a short repeat sequence found in 40% of sequenced bacteria and 90% of sequenced archaea. It is homologous to phages and plasmids and provides resistance against phages by targeting foreign homologous DNA, thus affecting plasmid conjugation. It is a part of the immune system in prokaryotes. CRISPRdigger tool was used to predict CRISPR in the sample genome. CRISPRdigger is highly accurate and can predict the short palindromic repeat structures within CRISPR. The results can be found in "03.Genome_Component.tar.gz" or "03.Genome_Component" under the file named "CRISPR".

Gene function analysis

Gene function annotation

There are several commonly used functional databases for annotation, including GO (Gene Ontology), KEGG (Kyoto Encyclopedia of Genes and Genomes), COG (Cluster of Orthologous Groups of proteins), KOG, eggNOG (evolutionary genealogy of genes: Non-supervised Orthologous Groups), NR (Non-Redundant Protein Database), TCDB (Transporter Classification Database), Swiss-Prot, and CAZy (Carbohydrate-Active enZymes Database). The results can be found in "04.Genome_Function.tar.gz" or "04.Genome_Function/General_Gene_Annotation" for each database.

The basic steps for functional annotation are as follows:

Perform a BLAST comparison (blastp) between the predicted genes and each functional database (with an e-value threshold of $\leq 1e-5$).

Filter the BLAST results: For each sequence, select the highest-scoring alignment result (default criteria: identity $\geq 40\%$, coverage $\geq 40\%$) for annotation.

Effector

Secretory proteins

Secretory proteins are proteins that are synthesized inside cells and then secreted outside the cells to exert their functions, guided by signal peptides. Many important enzymes necessary for life activities are found in secretory proteins. The N-terminal of secretory proteins is a signal peptide composed of 15-30 amino acids, which plays a key role in protein secretion. The SignalP tool can be used for signal peptide prediction to detect the presence of a signal peptide and transmembrane regions, and to predict whether a protein sequence is a secretory protein. The results can be found in "04.Genome_Function.tar.gz" or "04.Genome_Function/*/Effector/Secretory_Protein".

Prediction of Secretion System Proteins and T3SS Effector Proteins

Pathogenic bacteria secrete proteins into the extracellular space or host cells through the Type N Secretion Systems (TNSS). Currently, seven types of TNSS have been identified (Type I-VII). These proteins control immune responses, cell apoptosis, and pathological reactions. Among them, the T3SS of Gram-negative bacteria is commonly studied at the molecular level to understand the pathogenicity, infection mechanisms, and virulence effects, making it a well-researched secretion system.

For the TNSS system, secretion system-related proteins are extracted from the protein sequence functional database annotation results for annotation. The results can be found in "04.Genome_Function.tar.gz" or "04.Genome_Function/*/Effector/TnSS".

For Gram-negative bacteria, the EffectiveT3 software is used to predict T3SS effector proteins. The results can be found in "04.Genome_Function.tar.gz" or "04.Genome_Function/*/Effector/T3SS".

Analysis of secondary metabolism gene clusters

Secondary metabolites are synthesized by microorganisms from primary metabolites during a specific growth phase. They do not have a clear function in the microbial life activities and are not essential for growth and reproduction. The antiSMASH program (version 2.0.2) was used to predict the genome for secondary metabolite gene clusters.

Results can be found in "04.Genome_Function.tar.gz" or "04.Genome_Function/*/Effector/Secondary_Metabolism".

Toxicity or Pathogenicity Analysis

For bacteria, there are currently annotated pathogenicity and resistance databases available for analysis. These include the Pathogen-Host Interactions Database (PHI), Virulence Factors of Pathogenic Bacteria (VFDB), Antibiotic Resistance Genes Database (ARDB), and Comprehensive Antibiotic Research Database (CARD). The results can be found in "04.Genome_Function.tar.gz" or the respective databases within the "04.Genome_Function/Pathogenicity" folder.

Whole-genome collinearity analysis

Collinearity refers to the gene linkage relationship in genetics, which is the phenomenon where homologous genes on different species' genomes are arranged in the same order. The degree of collinearity between two species can serve as a scale to measure their evolutionary distance and determine their phylogenetic relationship. Results can be found in "06.Comparative_Genomics.tar.gz" or "Synteny" in "06.Comparative_Genomics".

SNP Analysis

SNP (Single Nucleotide Polymorphism) refers to DNA sequence polymorphism caused by single nucleotide variations at the genomic level, including single base substitutions, insertions, and deletions.

MUMmer software (version 3.22) was used for individual SNP detection, and SNP functionality was annotated based on the positional relationship and interaction between SNPs and genes. Results can be found in "06.Comparative_Genomics.tar.gz" or "SNP" in "06.Comparative_Genomics".

InDel Analysis

InDel refers to the insertion and deletion of small fragments in the genome. The LASTZ software was used to detect small fragment insertions and deletions (InDels) with a length of less than 50bp. The results can be found in "06.Comparative_Genomics.tar.gz" or "InDel" in "06.Comparative_Genomics".

Structural Variation (SV) refers to large-scale insertions, deletions, inversions, translocations, and other sequence changes at the genomic level.

To analyze SV, the MUMmer software (version 3.22) is used for fast alignment of two genomes to determine the relative orientation of sequences and adjust the genome by aligning the starting points. The Lastz software (version 1.02.00) is used for whole-genome alignment. Software packages like chainNet are used to connect scattered alignment results into longer alignment results in formats such as chain and net. Based on the new arrangement of alignment blocks and their relative orientations, alignment blocks representing translocations, inversions, and translocations with inversions are identified. The SV regions between adjacent alignment blocks on the two genomes are determined based on their distance relationships.

The results can be found in "06.Comparative_Genomics.tar.gz" or "SV" in "06.Comparative_Genomics".

Population Evolution Analysis

Analysis of Shared and Specific Genes

Genes that are present in all samples are referred to as core genes, while genes that are not shared among all samples are called dispensable genes. Specific genes are genes that are unique to a particular sample. The combination of all dispensable genes and core genes is referred to as the pan-gene. The core genes and specific genes are likely to correspond to common characteristics and specific traits of the samples, and can be used as a basis for studying functional differences between samples.

The results can be found in "06.Comparative_Genomics.tar.gz" or "Core-Pan" in "06.Comparative_Genomics".

In genomic evolution, a gene can produce two or more copies through gene duplication, which then form a gene family. A gene family is a group of genes that share a common ancestor, and the different genes within the family often have similar structures and functions. Since gene families are formed through species divergence, studying gene family evolution can reveal the evolutionary history of genes, gene functions, and species divergence, providing evidence and clues for predicting unknown gene functions and studying gene evolutionary history.

To analyze gene families, the BLAST software is used to pairwise compare the proteins of multiple target genomes. The unreliable alignment results are filtered, and then redundancy is removed using Solar. Hcluster-sg is used to cluster proteins based on their alignment similarities, resulting in gene family clustering. The results can be found in "06.Comparative_Genomics.tar.gz" or "Gene_Family" in "06.Comparative_Genomics".

Species Evolution Analysis

A phylogenetic tree, also known as an evolutionary tree, is a tree that depicts the evolutionary relationships between species, indicating their common ancestors. It is used to represent the results of phylogenetic analysis and describe the evolutionary relationships between species.

To analyze species evolution, the Neighbor-Joining (NJ) method is used to construct a phylogenetic tree. The results can be found in "06.Comparative_Genomics.tar.gz" or "Phylogenetic_Tree" in "06.Comparative_Genomics".

Novogene

Methylation Analysis

Using SMRT Link v8.0.0, we perform methylation site detection and predict nucleotide motifs for potential methylation transferase recognition on the final genome assembly results. The predicted modification types include 6mA, 4mC, 5mC, as well as unknown types (modified_base).

Genome Visualization Analysis

Whole-genome Map

For the assembled genome sequence of the sequencing sample, combined with the analysis of the sample genome composition and functional annotation from various databases, the Circos software is used to visually display the sample genome.

The results can be found in "05.Genome_Drawing.tar.gz" or "Genome_Overview" within the "05.Genome_Drawing" directory.